MIL-A-27471B(USAF)
13 October 1969
Superseding
MIL-A-27471A(USAF)
12 November 1964

# MILITARY SPECIFICATION

## ADAPTER, PRESSURE-REDUCER, IN-LINE CRU-43A/A

### 1. SCOPE

1.1 This specification covers one type of oxygen adapter designated CRU-43A/A.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

| | |
|---|---|
| BB-N-411 | Nitrogen, Technical |

Military

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of |
| MIL-D-1000 | Drawing, Engineering and Associated Lists |
| MIL-C-5040 | Cord, Nylon |
| MIL-O-27210 | Oxygen, Aviator's Breathing, Liquid and Gas |
| MIL-M-27274 | Mask, Oxygen, MBU-5/P |
| MIL-O-27335 | Oxygen System, Survival Container - Oxygen Kit, General Specification for |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-595 | Colors |

FSC 1660

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Military

| | |
|---|---|
| MIL-STD-130 | Identification Marking of U.S. Military Property |
| MIL-STD-143 | Specifications and Standards, Order of Precedence for the Selection of |
| MIL-STD-470 | Maintainability Program Requirements (for Systems and Equipments) |
| MIL-STD-471 | Maintainability Demonstration |
| MIL-STD-721 | Definition of Effectiveness Terms for Reliability, Maintainability, Human Factors, and Safety |
| MIL-STD-781 | Reliability Tests, Exponential Distribution |
| MIL-STD-794 | Parts and Equipment, Procedure for Packaging and Packing of |
| MIL-STD-831 | Test Reports, Preparation of |
| MS27796 | Connector - Bayonet, Three Pin, Oxygen Mask |
| MS29513 | Packing, Preformed, Hydrocarbon Fuel Resistant, "O" Ring |
| MS33586 | Metals, Definition of Dissimilar |

DRAWINGS

Air Force

| | |
|---|---|
| 57B3657 | Plate - Mounting of Connector, Oxygen Mask to Regulator, Female Assembly of |
| 57B3661 | Connector-Bayonet, Three Prongs, Oxygen Mask Assembly of |

(Copies of specifications, standards, drawings and publications required by suppliers in connection with the specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society of Mechanical Engineers

| | |
|---|---|
| USAS 46.1 | Surface Texture |

(Applications for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, N.Y. 10017.)



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## 3. REQUIREMENTS

*3.1 QUALIFICATION. Adapters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.2 SELECTION OF SPECIFICATIONS AND STANDARDS. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 MATERIALS. Any material that deteriorates easily or is otherwise easily affected adversely by continued service use with oxygen shall not be used. The materials shall be entirely suitable for the purpose intended. Materials that will come in contact with oxygen shall be entirely free from oil, grease, or any other combustible materials.

3.3.1 NONFERROUS MATERIALS. Except where ferrous materials are essential, nonferrous materials shall be used for all parts of the adapter.

3.3.2 METALS. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion caused by fuels, salt spray, or any other atmospheric conditions likely to be met in storage or normal service.

3.3.2.1 DISSIMILAR METALS. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.3.3 ELASTOMERS. All elastomers used shall have been manufactured within 12 months of the date the adapter is manufactured. The cure date (quarter and year) shall be stamped on the elastomer component, provided there is sufficient area. The figures shall be at least 3/16 inch high. Cure dates and age limitations shall not apply to silicone. All elastomers shall be free from foreign agents that might cause objectionable odors.

3.3.4 PROTECTIVE TREATMENT. When materials are used in the construction of the adapter that are subject to corrosion in oxygen, salt air, or any other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.4 DESIGN. The adapter shall be designed to adapt high altitude seat kit oxygen system (MIL-O-27335) to the MBU-5/P pressure breathing oxygen mask specified in MIL-M-27274. The design of the adapter shall include a pressure reduction mechanism to insure correct mask breathing pressure.



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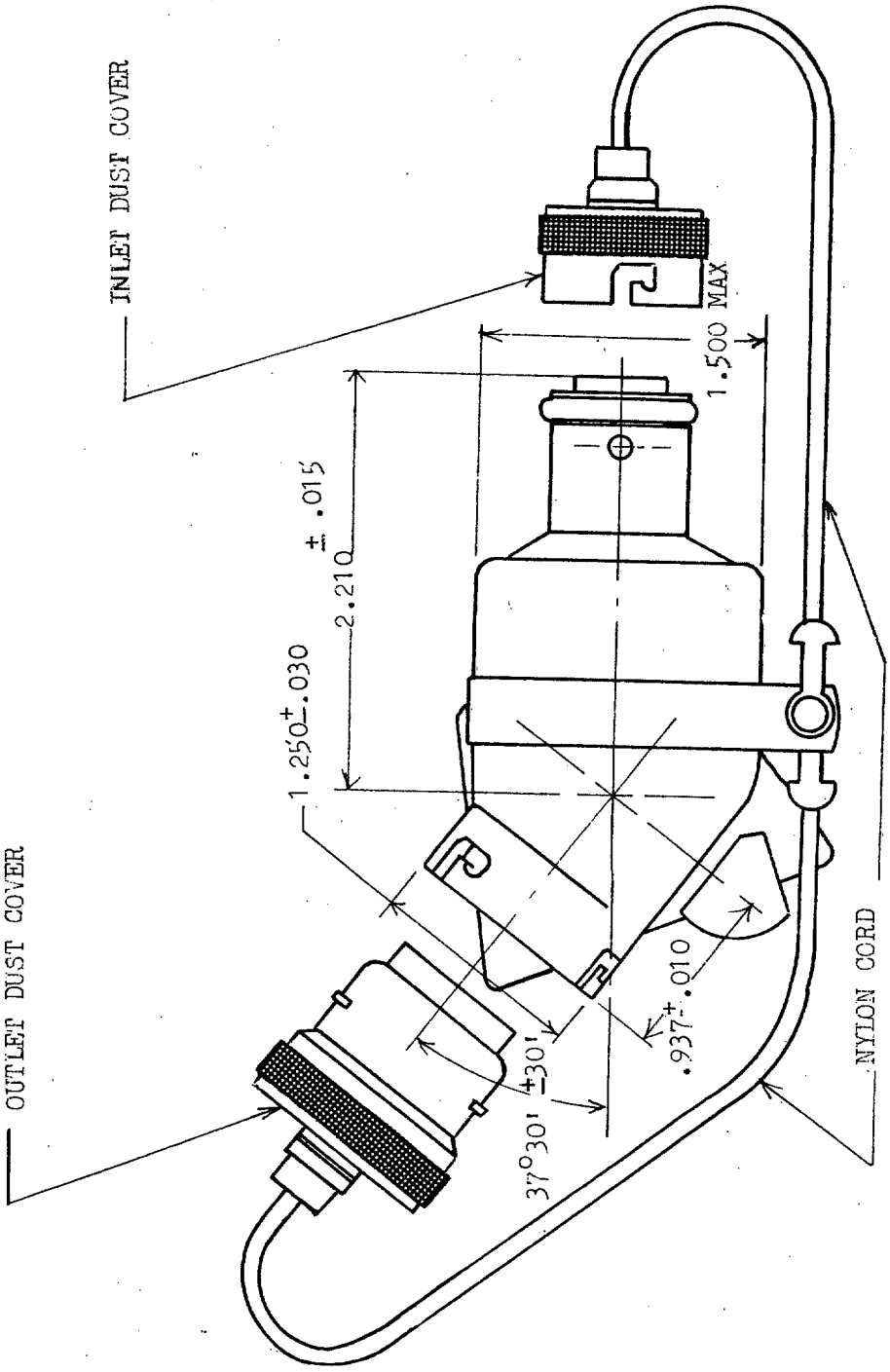
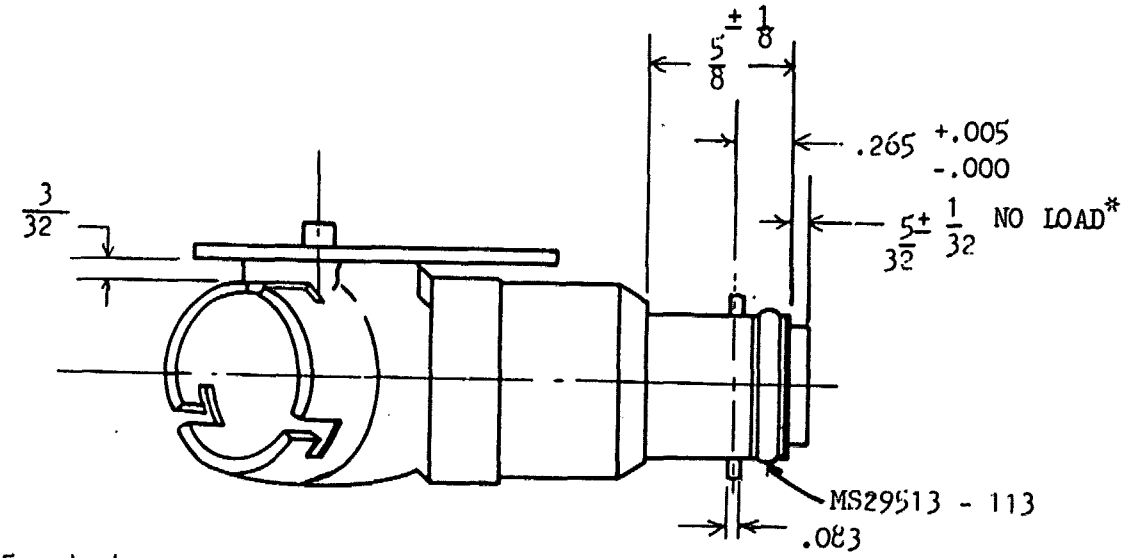
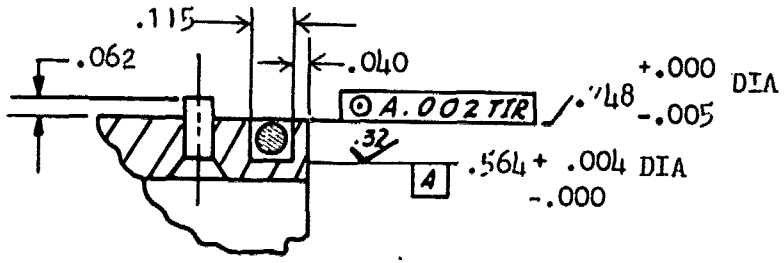



FIGURE 1 - ADAPTER



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ENLARGED SECTION
AT MALE END

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE INCHES AND TOLERANCES ARE:

| FRACTIONS | DECIMALS | ANGLES |
|---|---|---|
| ± 1/64 | ±.005 | ± 1/2° |

* LOAD TO DEPRESS SLEEVE FLUSH WITH END OF PORT SHALL BE 5 +1 POUND

ALL SURFACES 63 FINISH PER USAS 46.1 UNLESS OTHERWISE NOTED.

FIGURE 2 - ADAPTER INLET





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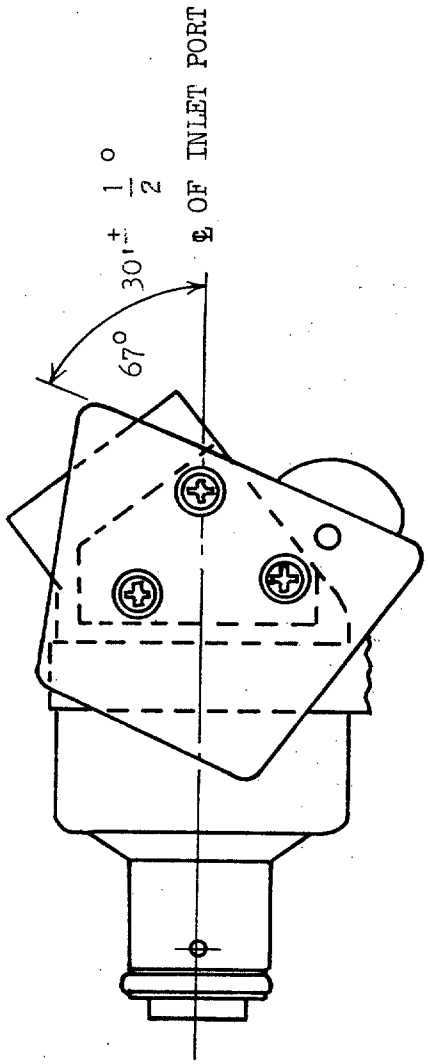



FIGURE 3 - PLATE ALIGNMENT



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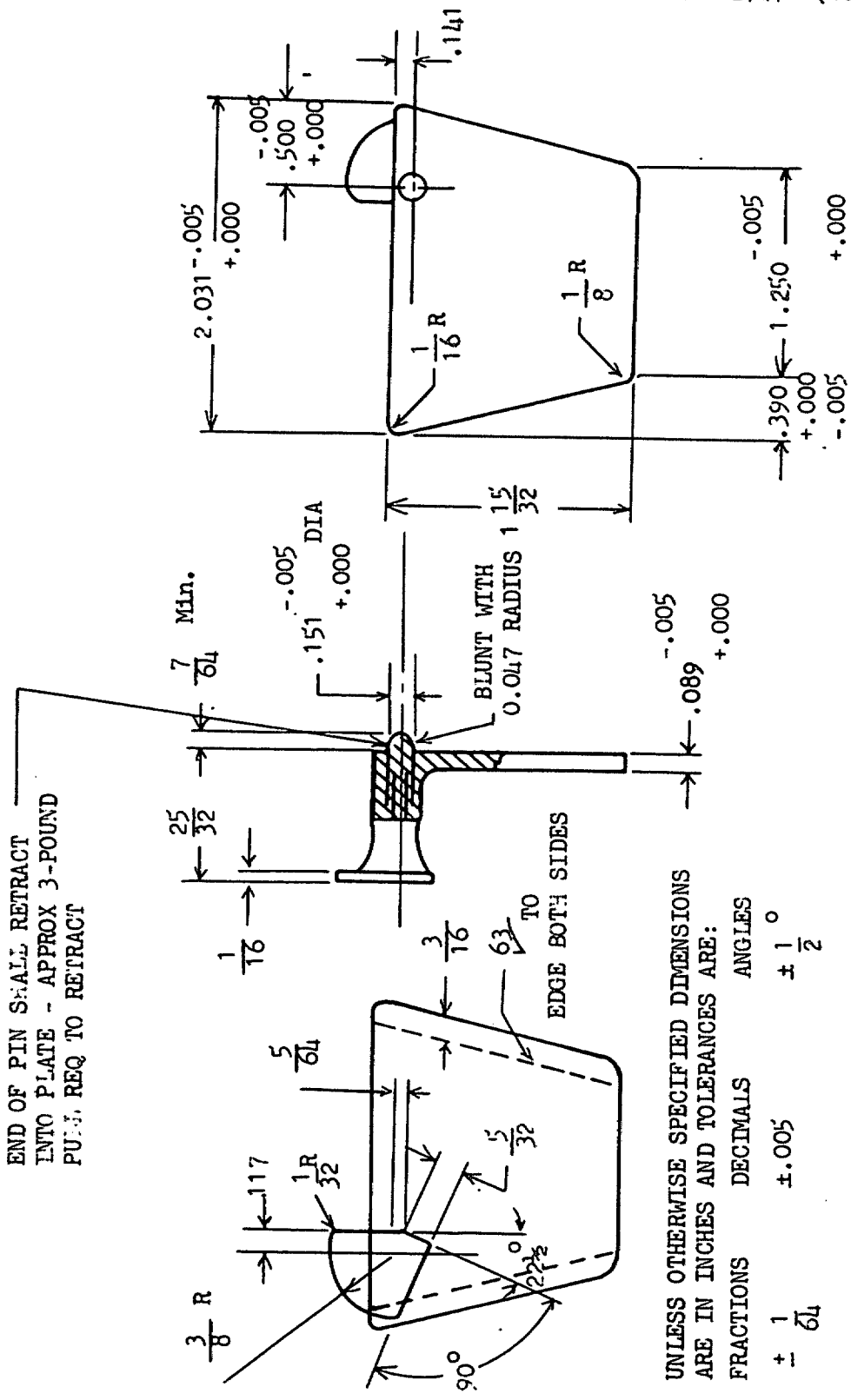
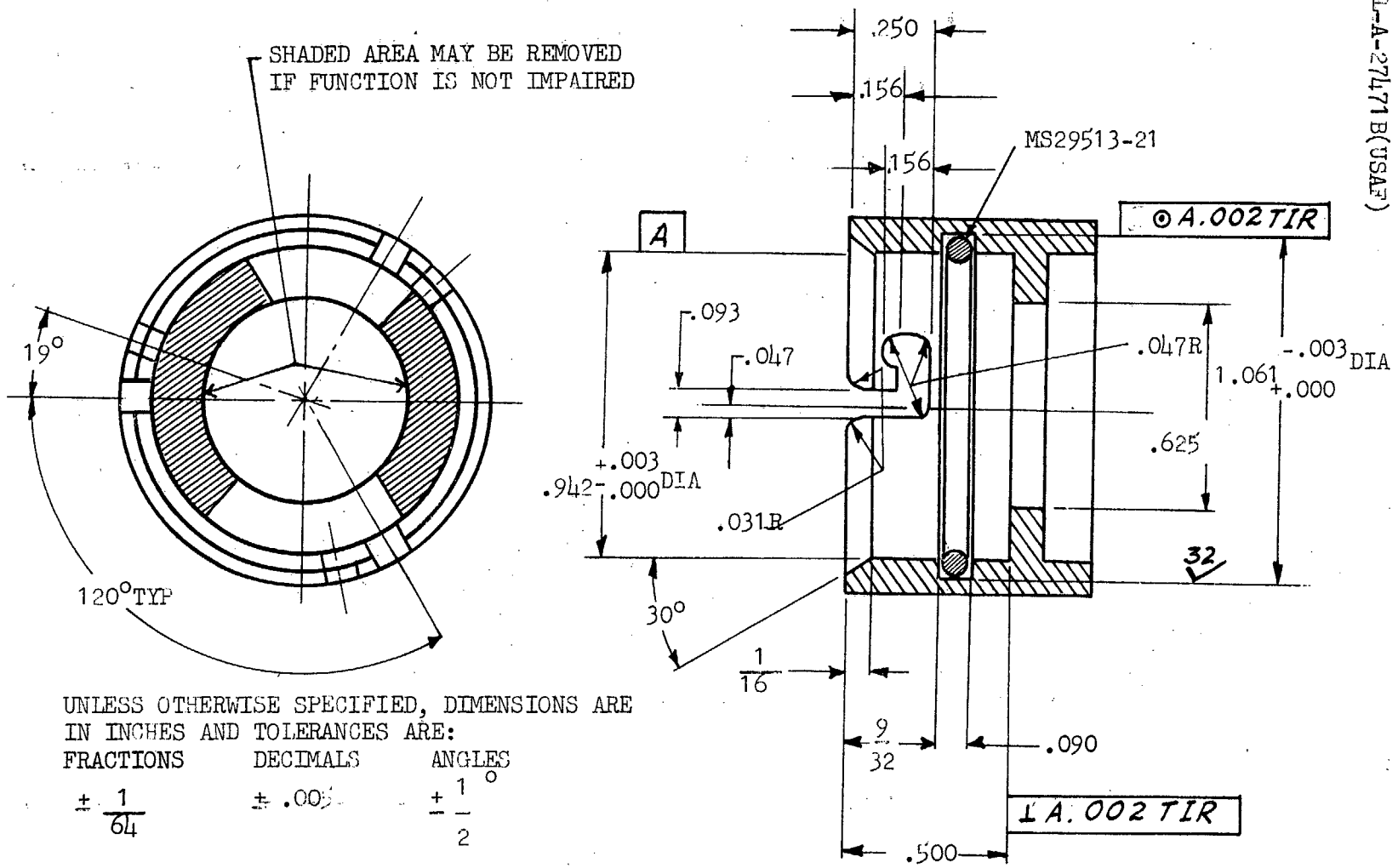



PLATE SHALL MATE SMOOTHLY WITH AF DRAWING 57B3657

FIGURE 4 - PLATE AND LOCKING PIN ASSEMBLY



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UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES AND TOLERANCES ARE:

| FRACTIONS | DECIMALS | ANGLES |
|---|---|---|
| $\pm \frac{1}{64}$ | ± .00[illegible] | $\pm \frac{1}{2}^{\circ}$ |

ALL SURFACES 63 FINISH PER USAS 46.1 UNLESS OTHERWISE NOTED.

OUTLET SHALL MATE PROPERLY WITH MS27796.

FIGURE 5 - ADAPTER OUTLET



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The adapter shall conform to figures 1 through 5. The design shall include a safety feature to provide breathing pressures for all rates of flow at altitudes above 30,000 feet.

3.5 CONSTRUCTION. The adapter shall be constructed so that no parts will become loose in service. It shall be built to withstand the strains, jars, vibrations, and any other conditions incident to shipment, storage, installation, and service.

*3.5.1 INLET AND OUTLET DUST COVERS. The inlet and outlet of the adapter shall have covers to prevent the entry of dust and foreign material when not in use. The dust covers shall provide a smooth and positive lock with the parts. The dust covers shall be attached to the adapter body by means of a nylon cord, Type II of MIL-C-5040. The nylon cord shall correspond to color 34127 of FED-STD-595.

*3.5.2 DOVETAIL PLATE. The adapter shall have a male dovetail plate conforming to figures 1, 3, and 4. The plate shall have a manual lock and a release that are readily accessible for use. The plate shall mate smoothly with AF Drawing 57B3657.

3.6 PART NUMBERING OF INTERCHANGEABLE PARTS. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-1000 shall govern the manufacturer's part numbers and changes thereto.

3.7 PERFORMANCE

3.7.1 PRESSURE REDUCTION CHARACTERISTICS. The adapter shall meet the pressure reduction characteristics specified in Table I.

3.7.2 OVERPRESSURE. The adapter shall withstand a pressure of 50 inches of water without leakage except for altitude compensating ports which shall not exceed 100 cubic centimeters per minute.

3.7.3 LOW TEMPERATURE OPERATION. While subjected to an inlet pressure of 3.5 inches (plus or minus 0.05 inch) of water, the adapter shall operate satisfactorily at minus 65 degrees (plus or minus 2 degrees) Fahrenheit. Upon conclusion of this test and while still at minus 65 degrees (plus or minus 2 degrees) Fahrenheit, the adapter shall meet the requirements of 3.7.1.

3.7.4 HIGH TEMPERATURE OPERATION. While subjected to an inlet pressure of 3.5 inches (plus or minus 0.05 inch) of water, the adapter shall operate satisfactorily at plus 160 degrees (plus or minus 2 degrees) Fahrenheit. Upon conclusion of this test and while still at plus 160 degrees (plus or minus 2 degrees) Fahrenheit, the adapter shall meet the requirements of 3.7.1.



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TABLE I - PRESSURE REDUCTION CHARACTERISTICS

| Flow Range of Oxygen (Liters Per Minute) | Adapter Inlet Pressure (Inches of Water) | | Adapter Outlet Pressure Range (Inches of Water) | Altitude Range Per 1,000 Feet |
|---|---|---|---|---|
| | Minimum | Maximum | | |
| 0 | +3.0 | +4.5 | -1.00 thru +3.0 | 0 thru 29 |
| 1 thru 10 | +2.6 | +3.6 | -1.35 thru +2.0 | 0 thru 29 |
| 11 thru 30 | +2.0 | +4.5 | -1.40 thru +2.5 | 0 thru 29 |
| 31 thru 50 | +2.4 | +4.9 | -1.50 thru +2.9 | 0 thru 29 |
| 51 thru 85 | +3.2 | +5.7 | -1.75 thru +3.5 | 0 thru 29 |
| 86 thru 90 | +4.2 | +6.7 | -2.00 thru +4.0 | 10 thru 29 |
| 0 | +3.0 | +4.5 | +0.5 thru +4.0 | 30 thru 37 |
| 0 | +3.0 | +4.5 | +0.8 thru +4.0 | 38 thru 40 |
| 0 | +4.5 | +11.5 | +2.0 thru +7.7 | 41 |
| 0 | +7.7 | +15.2 | +3.6 thru +9.1 | 42 |
| 0 | +10.9 | +17.9 | +5.20 thru +10.7 | 43 |
| 0 | +21.9 | +29.0 | +10.7 thru +15.8 | 47 |
| 1 thru 10 | +2.5 | +3.5 | +0.3 thru +2.8 | 30 thru 39 |
| 1 thru 10 | +2.5 | +3.6 | +0.3 thru +3.2 | 40 |
| 1 thru 10 | +4.5 | +8.5 | +2.0 thru +7.2 | 41 |
| 1 thru 10 | +6.9 | +11.7 | +3.2 thru +8.6 | 42 |
| 1 thru 10 | +10.9 | +14.8 | +5.2 thru +10.2 | 43 |
| 1 thru 10 | +22.7 | +25.8 | +11.1 thru +15.3 | 47 |
| 11 thru 30 | +2.0 | +4.5 | +0.60 thru +2.9 | 30 thru 39 |
| 11 thru 30 | +2.0 | +4.5 | +0.60 thru +4.4 | 40 |
| 11 thru 30 | +4.8 | +11.5 | +2.1 thru +7.1 | 41 |
| 11 thru 30 | +7.8 | +15.0 | +3.5 thru +8.5 | 42 |
| 11 thru 30 | +11.4 | +17.9 | +5.4 thru +10.1 | 43 |
| 11 thru 30 | +23.5 | +30.0 | +11.3 thru +15.2 | 47 |
| 31 thru 50 | +2.6 | +4.9 | +0.6 thru +3.0 | 30 thru 39 |
| 31 thru 50 | +2.4 | +4.9 | +0.9 thru +4.5 | 40 |
| 31 thru 50 | +4.7 | +11.9 | +2.0 thru +7.2 | 41 |
| 31 thru 50 | +7.5 | +15.6 | +3.40 thru +8.6 | 42 |
| 31 thru 50 | +11.3 | +18.3 | +5.30 thru +10.2 | 43 |
| 31 thru 50 | +23.5 | +29.4 | +11.20 thru +15.3 | 47 |
| 51 thru 70 | +4.2 | +5.7 | +1.0 thru +3.3 | 30 thru 39 |
| 51 thru 70 | +4.8 | +5.7 | +1.3 thru +4.8 | 40 |
| 51 thru 70 | +5.5 | +12.7 | +2.2 thru +7.4 | 41 |
| 51 thru 70 | +8.2 | +16.3 | +3.6 thru +8.8 | 42 |
| 51 thru 70 | +12.0 | +19.0 | +5.5 thru +10.4 | 43 |
| 51 thru 70 | +24.0 | +30.1 | +11.4 thru +15.5 | 47 |
| 71 thru 90 | +4.2 | +6.7 | +0.90 thru +3.6 | 30 thru 39 |
| 71 thru 90 | +5.0 | +6.7 | +1.20 thru +4.9 | 40 |
| 71 thru 90 | +4.0 | +13.5 | +1.80 thru +7.3 | 41 |
| 71 thru 90 | +7.2 | +17.2 | +3.20 thru +8.7 | 42 |
| 71 thru 90 | +11.5 | +19.9 | +5.15 thru +10.3 | 43 |
| 71 thru 90 | +24.5 | +31.0 | +11.00 thru +15.4 | 47 |



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3.7.5 LOW TEMPERATURE EXPOSURE. After continuous exposure to a temperature of minus 85 degrees (plus or minus 5 degrees) Fahrenheit for 48 hours, the adapter shall pass the tests specified in 4.5.1.

3.7.6 HIGH TEMPERATURE EXPOSURE. After continuous exposure to a temperature of not less than plus 160 degrees Fahrenheit for 7 hours, the adapter shall pass the tests specified in 4.5.1.

3.7.7 ALTITUDE CYCLING. At no flow rate or inlet pressure, the adapter shall be cycled 100 times at an altitude from 0 feet to 50,000 feet and back to 0 feet in not less than 30 minutes. The adapter shall then meet the tests specified in 4.5.1.

3.7.8 G FORCE. When subjected to forces of 6g, the adapter shall operate as specified herein. After being subjected to a force of 22g for a period of 0.01 second, the adapter shall meet the tests specified in 4.5.1.

3.7.9 ORIENTATION. The adapter shall operate as specified herein when tested in vertical and horizontal positions.

3.7.10 RELIABILITY. The adapter shall have a minimum mean time between failure (MTBF) of 870 hours with a minimum reliability of 0.9816 for a 16-hour mission at a confidence level of 0.90.

*3.7.10.1 LONGEVITY. All reparable adapters shall have a minimum operating life span (equipment longevity as defined in MIL-STD-781) of not less than 2700 hours before wearout failures occur or the equipment consistently fails to meet the specified MTBF index. Parts requiring replacement within this period of time and the normal operating period of the parts requiring replacement shall be reported by the contractor to the procuring activity.

*3.7.11 MAINTAINABILITY. Maintainability Program requirements shall be as described in MIL-STD-470.

*3.7.11.1 PREVENTIVE MAINTENANCE. Preventive maintenance requirements shall be kept at a minimum by using (but not limiting to):

a. Fail-safe features
b. Components requiring little or no preventive maintenance.

*3.7.11.2 DOWNTIME. The total mean active downtime (as defined in MIL-STD-721) in the maintenance of the adapter shall be the minimum possible by designing for (but not limiting to):

a. Rapid and positive detection of malfunction.
b. Ease of fault correction.
c. Rapid and positive verification of correction.



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3.8 WEIGHT. The adapter shall weigh not more than 7 ounces.

3.9 IDENTIFICATION OF PRODUCT. Equipment, assemblies, and parts shall be marked in accordance with MIL-STD-130.

3.9.1 DATE. The date of manufacture, consisting of quarter and year, shall be legibly stamped with ink in a place where it will not be defaced because of usage. A suitable rubber stamp made in the form of a 1/2-inch square with a horizontal line dividing the square into two segments shall be used for this marking. The upper segment shall contain the number of the quarter, and the lower segment shall contain the last two numbers of the year.

3.10 WORKMANSHIP. The adapters shall be manufactured in a thoroughly workmanlike manner. The finished adapters and all integral parts shall be clean and free of oils and any other materials that might adversely affect the operation of the adapter.

4. QUALITY ASSURANCE PROVISIONS.

*4.1 RESPONSIBILITY FOR INSPECTION. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 CLASSIFICATION OF INSPECTIONS. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 TEST CONDITIONS

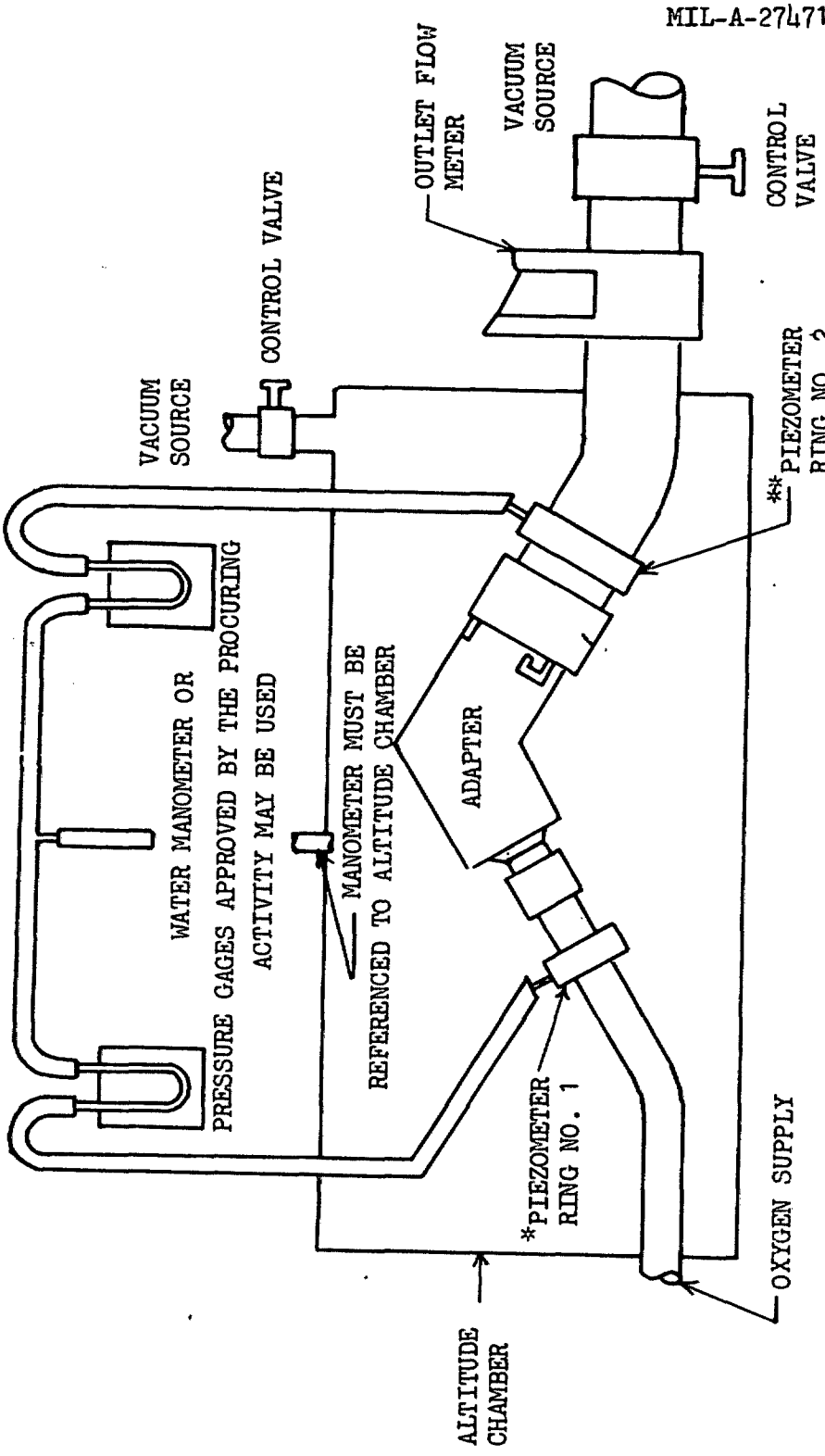
4.3.1 GAS. The high pressure gas used in testing the adapters shall be oxygen conforming to type I of MIL-O-27210; water-pumped nitrogen conforming to type I, class I, grade B of BB-N-411; or water-pumped air equivalent in dryness to BB-N-411 nitrogen. If nitrogen or air is used, appropriate density corrective factors shall be applied to the flow meter and to the performance of the adapters.

4.3.2 PRESSURE AND TEMPERATURE. Unless otherwise specified, the test conditions of pressure and temperature will be 29.92 inches of mercury and 77 degrees (plus or minus 10 degrees) Fahrenheit, respectively. For any other conditions of pressure and temperature, appropriate corrections shall be made to the test results.

4.3.3 TEST SET-UP. Where applicable, the test set-up shown in Figure 6 shall be used for tests.



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* PIEZOMETER RING NO. 1 MAY BE CONNECTED TO AIRLOCK, INC, FITTING NO. 265

** PIEZOMETER RING NO. 2 MAY BE CONNECTED TO A FITTING IN CONFORMANCE TO 57B3661

FIGURE 6 - TEST SETUP FOR PRESSURE REDUCTION CHARACTERISTICS



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*4.4 QUALIFICATION INSPECTION

*4.4.1 TEST SAMPLES. The test samples shall consist of four adapters representative of the production equipment. Two of the samples shall be subjected to all the tests of this specification, and four samples (to include the two subjected to all tests) shall be subjected to the qualification reliability tests of 4.4.4. The samples shall be identified with the manufacturer's part number and any other information required by the qualifying activity.

*4.4.2 TEST REPORT. When the tests are conducted at a location other than the laboratory of the qualifying activity, two copies of a test report in accordance with MIL-STD-831 and two test samples shall be furnished the qualifying activity.

*4.4.3 QUALIFICATION TESTS. The qualification tests shall consist of all tests specified under 4.6. Unless otherwise specified by the procuring activity, the tests shall be conducted in the order listed.

| | | |
|---|---|---|
| a. | Examination of product | 4.6.1 |
| b. | Pressure reduction characteristics | 4.6.2 |
| c. | Overpressure | 4.6.3 |
| d. | Low temperature operation | 4.6.4 |
| e. | High temperature operation | 4.6.5 |
| f. | Low temperature exposure | 4.6.6 |
| g. | High temperature exposure | 4.6.7 |
| h. | Altitude cycling | 4.6.8 |
| i. | G forces | 4.6.9 |
| | (1) Operating Conditions | 4.6.9.1 |
| | (2) Nonoperating Conditions | 4.6.9.2 |
| j. | Orientation | 4.6.10 |
| k. | Reliability | 4.6.11 |
| | (1) Accept - Reject Criteria | 4.6.11.1 |
| l. | Maintainability verification | 4.6.12 |
| m. | Preservation and Packaging | 4.6.13 |



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*4.4.4 QUALIFICATION RELIABILITY TESTS. Four adapters shall be tested by the contractor for reliability in accordance with 4.6.11 and MIL-STD-781. If required to meet the longevity requirements of 3.7.10.1 (reparable items only), two adapters shall be continued through a longevity test in accordance with MIL-STD-781. Duration of the reliability test (MTBF demonstration) and acceptance shall be judged on the basis of the table listed in 4.6.11.1.

*4.5 QUALITY CONFORMANCE INSPECTION: Quality conformance inspection shall consist of:

a. Individual tests
b. Sampling tests

4.5.1 INDIVIDUAL TESTS. Each adapter shall be subjected to the following tests as described under 4.6:

a. Examination of product
b. Pressure reduction characteristics (at ground level only)
c. Overpressure.

4.5.2 SAMPLING TESTS

4.5.2.1 SAMPLING PLAN A. One adapter shall be selected at random from each lot of 100 adapters (or fraction thereof) and shall be subjected to the sampling Plan A tests listed in Table II.

4.5.2.2 SAMPLING PLAN B. Two adapters shall be selected at random from each lot of 500 adapters (or fraction thereof) and shall be subjected to the sampling plan B tests listed in Table II. After completion of these tests, the adapters shall pass the tests specified in 4.5.1.

4.5.2.3 SAMPLING PLAN C. Four adapters shall be selected at random from each lot of 1,000 adapters (or fraction thereof) and shall be subjected to the sampling plan C test of Table II.

TABLE II - Sampling Plan

| Sampling Plan A | 4.5.2.1 |
|---|---|
| Inspection | Method |
| Low Temp. Operation | 4.6.4 |
| High Temp Operation | 4.6.5 |
| Pressure Reduction Characteristics (at all altitudes) | 4.6.2 |
| **Sampling Plan B** | **4.5.2.2** |
| Low Temp. Exposure | 4.6.6 |
| High Temp. Exposure | 4.6.7 |
| **Sampling Plan C** | **4.5.2.3** |
| Reliability | 4.6.11 |



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4.5.2.4 REJECTION AND RETEST. When one or more items from a lot fail to meet the specification, acceptance of all items in the lot shall be withheld until the extent and cause of failure have been determined. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the sampling tests shall be repeated.

4.5.2.5 INDIVIDUAL TESTS MAY CONTINUE. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot or lots produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 DEFECTS IN ITEMS ALREADY ACCEPTED. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.5.4 VIBRATION AND NOISES. Vibration, flutter, or chatter occurring during breathing or simulated breathing tests shall be cause for rejection of the adapter. If this occurs on a sampling test representing a lot of adapters, the entire lot shall be rejected. If the adapter vibrates during continuous flow testing, but the vibration can be stopped by altering the flow or inlet pressures and then returning to the test point in question, the adapter shall be acceptable providing the entire test is completed within the limitations precribed for that test. If the vibration persists after three attempts at a single test point, the adapter shall not be acceptable.

4.6 TEST METHODS.

4.6.1 EXAMINATION OF PRODUCT. Unless otherwise specified by the procuring activity, each adapter shall be examined for conformance to this specification with respect to workmanship, weight, dimensions, and marking.

4.6.2 PRESSURE REDUCTION CHARACTERISTICS. The adapter shall be connected to a pressure source capable of producing the inlet pressures specified in Table III. The adapter outlet shall be connected to a vacuum source. When the inlet pressures are established in accordance with Table III, the adapter outlet shall be introduced to the vacuum supply in such a manner that the proper flow rates are established. Altitude requirements shall be simulated by placing the adapter in an appropriate environment. When respective inlet pressure (minimum and maximum) flow rates and altitude conditions are established, the outlet pressures, when measured as shown in figure 6, shall fall within the range of outlet pressures shown in Table III for all altitudes.

4.6.3 OVERPRESSURE. With the altitude compensating ports plugged and the outlet closed, a pressure of 50 inches of water shall be applied to the inlet; any outward leakage of the adapter shall be cause for rejection. With the altitude compensating ports unplugged and the adapter outlet closed, a pressure of 50 inches of water shall be applied to the inlet; outward leakage of the adapter exceeding 100 cubic centimeters per minute shall be cause for rejection.



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TABLE III
PRESSURE REDUCTION CHARACTERISTICS - TEST POINT

| Altitude Per 1,000 Feet | Flow of Oxygen (Liters Per Minute) | Adapter Inlet Pressure (Inches of Water) Minimum | Adapter Inlet Pressure (Inches of Water) Maximum | Adapter Outlet Pressure (Inches of Water) |
|---|---|---|---|---|
| ground level | 0 1/ | 3.0 | 4.5 | -1.0 thru +3.2 |
| | 10 | 2.6 | 3.6 | -1.35 " +2.0 |
| | 50 | 2.4 | 4.9 | -1.5 " +2.9 |
| | 85 | 3.2 | 5.7 | -1.75 " +3.5 |
| 30 | 0 1/ | 3.0 | 4.5 | +0.5 " +4.0 |
| | 10 | 2.5 | 3.5 | +0.3 " +2.8 |
| | 50 | 2.6 | 4.9 | +0.6 " +3.0 |
| | 90 | 4.2 | 6.7 | +0.9 " +3.6 |
| 41 | 0 1/ | 4.5 | 11.5 | +2.0 " +7.7 |
| | 10 | 4.5 | 8.5 | +2.0 " +7.2 |
| | 50 | 4.7 | 11.9 | +1.9 " +7.2 |
| | 90 | 4.0 | 13.5 | +1.4 " +7.3 |
| 47 | 0 1/ | 21.9 | 29.0 | +10.7 " +15.9 |
| | 10 | 22.7 | 25.8 | +11.1 " +15.3 |
| | 50 | 23.5 | 29.4 | +11.2 " +15.3 |
| | 90 | 24.5 | 31.0 | +11.0 " +15.7 |

1/ The outlet pressure for zero flow rates shall be maintained for one minute.

4.6.4 LOW TEMPERATURE OPERATION. With a test setup similar to the one specified in figure 6, the adapter shall be subjected to an inlet pressure of 3.5 inches (plus or minus 0.5 inch) of water and shall be stabilized at a temperature of minus 65 degrees (plus or minus 2 degrees) Fahrenheit. While under these conditions, the adapter shall be subjected to cyclic operation by changing outlet pressures so that flow rates will vary from 0 to 20 to 0 liters per minute. The cyclic operation shall consist of 15 (plus or minus 5) cycles per minute, each cycle given approximately equal time, and continuing for 50 cycles. Following the cyclic operation, and while the temperature is still stabilized at minus 65 degrees (plus or minus 2 degrees) Fahrenheit, the adapter shall be subjected to the test specified in 4.6.2. A 12 percent increase in outlet perssure is allowable for inlet pressures above 25 inches of water.

4.6.5 HIGH TEMPERATURE OPERATION. Except that the operating temperature shall be stabilized at plus 160 degrees (plus or minus 2 degrees) Fahrenheit, this test shall be the same as the test specified in 4.6.4.

4.6.6 LOW TEMPERATURE EXPOSURE. The adapter shall be subjected continuously for 48 hours to a temperature of minus 85 degrees (plus or minus 5 degrees) Fahrenheit. This exposure shall be at nonoperating conditions such as ambient inlet and outlet pressures and zero flow rates. After the 48-hour period of exposure, the adapter shall be stabilized to room temperature and subjected to the tests specified in 4.5.1.



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4.6.7 HIGH TEMPERATURE EXPOSURE. The adapter shall be subjected continuously for 7 hours to a temperature not less than plus 160 degrees Fahrenheit. This exposure shall be at nonoperating conditions such as ambient inlet and outlet pressures and zero flow rates. After the 7-hour period of exposure, the adapter shall be stabilized to room temperature and subjected to the tests specified in 4.5.1.

4.6.8 ALTITUDE CYCLING. At room temperature, zero flow rate, and ambient inlet and outlet pressures, the adapter shall be cycled 100 times in an altitude chamber or equivalent. Each cycle shall start at ground level, go up to 50,000 feet, and return to ground level. This test shall not be completed sooner than 30 minutes with each cycle given approximately equal time. Upon completion of this test, the adapter shall be subjected to the tests specified in 4.5.1.

4.6.9 G FORCES

4.6.9.1 OPERATING CONDITIONS. At ground level and at sufficient inlet pressures to cause flow rates for actual breathing exercises, the adapter shall be subjected to a force of up to 6g while actual or simulated breathing cycles are being conducted. After a sufficient number of breathing cycles, detection of any severe resistance to breathing shall be cause for rejection. Since the basis for acceptance or rejection under this test is personal opinion, the test shall show more than one opinion of the same nature before substantial grounds for rejection of the adapter can be advanced. Upon completion of this test, the adapter shall be subjected to the tests specified in 4.5.1.

4.6.9.2 NONOPERATING CONDITIONS. Under nonoperating conditions the adapter shall be mounted in a suitable device and shall be subjected to a g force of approximately 22g for 0.01 second. The adapter shall be appropriately protected against the g forces that will act if the adapter strikes the ground. Upon completion of this test, the adapter shall be subjected to the tests specified in 4.5.1.

4.6.10 ORIENTATION. The adapter shall be subjected to the test specified in 4.6.2. For this test, the adapter shall be oriented in three positions (directions of flow):

a. Vertical (flow direction - up)

b. Vertical (flow direction - down)

c. Horizontal.

4.6.11 RELIABILITY. The four adapters (ref. para. 4.5.2.3) shall be subjected to a simulated breathing test. The test shall be conducted at room temperature at a rate of between 10 and 20 breathing cycles per minute. Cyclic breathing of the adapters shall be accomplished with peak flow rates of 30 + 5 liters per minute for 90 percent of the cycles and 70 + 10 liters per minute for 10 percent of the cycles (inhalation). During 20 percent of the time, the adapters are subjected to breathing cycles, they shall be vibrated at a frequency



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ranging from 500 to 2500 cycles per minute at a double amplitude of not less than 0.018 inch nor more than 0.020 inch. The rate of change of frequency shall be logarithmic and such that a maximum of 15 minutes is required to proceed from 500 cycles per minute to 2500 cycles per minute back to 500 cycles per minute. When there is no provision for logarithmic cycling, a linear rate of change may be used. The adapter shall be vibrated one-third of the vibration time on the longitudinal axis, one-third of the vibration time on the horizontal axis, and one-third of the vibration time on the lateral axis. The adapter shall be checked at intervals not to exceed 24 hours during reliability testing to determine if a failure has occurred. A failure shall be identified as the inability of the adapter to meet the ground level test point for Pressure Reduction Characteristics in paragraph 4.6.2. No parts shall be replaced as preventive maintenance during reliability testing; however, failed parts may be replaced. Acceptance will be judged on the basis of the table in 4.6.11.1. Recording, data handling, and reporting procedures shall be in accordance with MIL-STD-781.

*4.6.11.1 Accept-Reject Criteria.

| Failures Permitted | Test Time - Hours | Ratio |
|---|---|---|
| 1 | 3383 | 3.889 |
| 2 | 4630 | 2.661 |
| 3 | 5812 | 2.227 |
| 4 | 6954 | 1.998 |

*4.6.12 MAINTAINABILITY VERIFICATION. A maintainability verification demonstration method as described in MIL-STD-471 shall be selected and submitted for the approval of the procuring activity.

*4.6.13 PRESERVATION AND PACKAGING. Unless otherwise specified in the purchase order or contract, the supplier is responsible for the performance of all inspections required as listed in Table III, MIL-P-116 for the method of preservation being accomplished.

5. PREPARATION FOR DELIVERY

*5.1 PRESERVATION AND PACKAGING. Unless otherwise specified by the procuring activity, the adapters shall be prepared for delivery in accordance with MIL-STD-794.

*5.1.1 LEVELS A OR B. Each adapter shall be preserved and packaged in accordance with Method IC of MIL-P-116 without the use of contact preservatives and without cleaning by immersion in solvent.

*5.1.2 LEVEL C. Each adapter shall be provided preservation which will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the suppliers commercial practice when such meets the requirements of this level.



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*5.2 PACKING. The required level of packaging (A, B or C) shall be specified by the procuring activity (see 6.2).

*5.2.1 LEVEL A. Each adapter preserved as specified in 5.1.1 shall be packaged in exterior type weather resistant containers. All containers shall be closed, sealed, and reinforced in accordance with the appendix of the applicable specification.

*5.2.2 LEVEL B. All adapters preserved as specified in 5.1.1 shall be packaged in domestic type exterior containers. All containers shall be closed and strapped in accordance with appendix thereto.

*5.2.3 LEVEL C. All packaged adapters which require overpacking for acceptance by the carrier, shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet the uniform freight classification rules or regulations of other common carriers as applicable to the mode of transportation.

*5.3 PRECAUTIONARY MARKING. The following precautionary marking shall appear on each unit package:

"CAUTION: DO NOT ALLOW OIL OF ANY KIND TO BE USED ON OR ABOUT OXYGEN ADAPTERS."

6. NOTES

*6.1 INTENDED USE. The adapter covered by this specification is intended for use in breathing oxygen pressurizing systems installed in aircraft for adapting high altitude seat kit oxygen to a standard pressure breathing oxygen mask.

6.2 ORDERING DATA. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Level of packaging and packing desired.

*6.3 PROVISIONS FOR QUALIFICATION. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma Air Materiel Area (OCNE), Tinker AFB, Oklahoma 73145, and information pertaining to qualification of products may be obtained from that activity.



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# SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

| SPECIFICATION | | |
|---|---|---|
| ORGANIZATION | | CITY AND STATE |
| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT<br>$ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

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